

MCA 2008 – EXAMINATION PAPER

1. All admission tickets to circus cost Rs.12 for adults and Rs. 6 for children. There were 500 tickets sold for Rs. 5,280. The number of children's tickets sold is _____ the number of adult's tickets sold.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
2. A retail business has determined that its net income in terms of x , the number of items sold is given by the expression $x^2 + x - 380$. The number of items that must be sold for the net income to be zero is _____ 10.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
3. The average test scores for MCA I Semester is 84. The average test scores for the boys in that class is 90. The average test scores for the girls in that class is 81. The number of boys is _____ the number of girls.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
4. After 2 women leave a party, there are twice as many men as women. Then 9 men leave and there are twice as many women as men. The number of people remaining in the party after the 9 men left is _____ 8.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
5. On a 50-question multiple choice test, 3 points were given for each correct answer and 1 point was deducted for each wrong answer. A student who answered all the questions on the test received 98 points. The number of questions he answered incorrectly is _____ 14.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
6. A 20-m ladder leaning against a vertical wall with the base of the ladder 10 m from the wall is pulled 2 m further from the wall, causing the top of the ladder to drop x m. Then x is _____ 2m.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
7. Jana purchased p pencils and Stalin purchased 3 more than half as many pencils as Jana. The number of pencils Stalin purchased is _____ $(p+6)/2$.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
8. Raghu has a flat square garden with perimeter of 20 m. David has a flat rectangular garden with perimeter of x m and length 1 m longer than the width. The area of Raghu's garden is _____ the area of David's garden.
 1) greater than 2) equal to
 3) less than 4) cannot be determined
9. $xy=18$ and $x+y=9$. x is _____ y .
 1) greater than 2) equal to
 3) less than 4) cannot be determined
10. The average of two numbers is $2x$. If one of the numbers is y , the other number must be
 1) $4x-y$ 2) $x+4y$
 3) $4x+y$ 4) $x-4y$
11. Which of the following equals the reciprocal of $x-1/y$ where $x-1/y \neq 0$?
 1) $x/xy-1$ 2) $y/x-1$
 3) $1/x-y$ 4) $y/(xy-1)$
12. How many litres of pure alcohol must be added to a 100 litre solution that is 20% alcohol in order to produce a solution that is 25% alcohol?
 1) $7/2$ 2) 5
 3) $39/4$ 4) $20/3$

13. A rectangular floor 18 m by 10 m is to be completely covered with carpeting that costs Rs. x per square yard (1 yard = 0.9 m). In terms of x , what is the cost of carpeting?
- 1) $20x$ 2) $28x$
3) $30x$ 4) $32x$
14. For many positive integers, a , is it true that $a^2 \leq 2a$?
- 1) 1 2) 2
3) 3 4) 4
15. If m men can do a work in r days, then the number of days taken by $(m+n)$ men to do it is
- 1) $\frac{m+n}{mr}$ 2) $\frac{m(m+n)}{r}$
3) $\frac{(m+n)r}{m}$ 4) $\frac{mr}{(m+n)}$
16. A, B, C together can finish a piece of work in 4 days. A alone can do it in 12 days and B alone in 18 days. How many days will be taken by C to do it alone?
- 1) 21 2) 16
3) 14 4) 9
17. A can complete a job in 9 days, B in 10 days and C in 15 days. B and C start the work and forced to leave after 2 days. The number of days taken by A to finish the remaining work is
- 1) 8 2) 9
3) 6 4) 14
18. 4 men and 6 women finish a job in 8 days, while 3 men and 7 women finish it in 10 days. In how many days will 10 women finish it?
- 1) 24 2) 32
3) 36 4) 40
19. Two pipes M and N fill a cistern in 24 min., and 32 min. respectively. If both the pipes are opened together, then after how many minutes N should be closed so that the tank is full in 18 minutes?
- 1) 6 2) 8
3) 10 4) 12
20. A man covers a certain distance in a scooter. Had he moved 3 kmph faster, he would have taken 40 min. less. If he had moved 2 kmph slower, he would have taken 40 min. more. The distance in km is
- 1) 20 2) 36
3) 37.5 4) 40
21. Walking $\frac{5}{6}$ of his usual speed, a man is 10 min late. The usual time taken by him to cover that distance is
- 1) 1 hour 2) 50 min
3) 12 min 4) 8 min 20 sec.
22. Two trains start at the same time from station Chennai and Mangalore and proceed towards each other at 36 kmph and 42 kmph respectively. When they meet, it is found that one train has traveled 48 km more than the other. The distance between the two stations (in km) is
- 1) 624 2) 636
3) 544 4) 460
23. A train 280 m long, running with a speed of 63 kmph will pass an electric pole in
- 1) 160 sec 2) 16 sec
3) 18 sec 4) 15 sec
24. Two trains whose lengths are 180 m and 220 m respectively are running in directions opposite to one another with respective speeds of 40 km/h and 50 km/h. Time taken by them in crossing one another will be
- 1) 16 sec 2) 17 sec
3) 18 sec 4) 22 sec
25. A train overtakes 2 persons who are walking in the same direction in which the train is going, at the rate of 2 km/h and 4 km/h and passes them completely in 9 and 10 seconds respectively. The length of the train (in metres) is
- 1) 45 2) 54
3) 50 4) 72
26. A man can row 5 km/h in still water. If the river is running at 1 km/h, it takes him 75 minutes to row to a place and back. How far is the place?
- 1) 2.5 km 2) 3 km
3) 4 km 4) 5 km

27. A man can swim 3 km/h in still water. If the velocity of the stream is 2 km/h, the time taken by him to swim to a place 10 km upstream and back, is

1) 8.5 hrs 2) 9.2 hrs
3) 10 hrs 4) 12 hrs

28. Consider an experiment in which two six sided dice are tossed. Find the probability that the sum is odd or more than 8.

1) $11/18$ 2) $17/36$
3) $9/18$ 4) $31/36$

29. In a history class that consisted of 30 students the number of seniors was 3 more than twice the number of juniors, and $3/10$ of the students were neither juniors nor seniors. The number of juniors in the class is

1) 11 2) 9
3) 6 4) 12

30. If a finite set S contains n elements, then the number of non-empty proper subsets of S is

1) $2 \cdot 2^{n-1}$ 2) $2(2^n - 1)$
3) $(2^n - 1)$ 4) $2(2^{n-1} - 1)$

31. If $X = \{4^n - 3n - 1 : n \in \mathbb{N}\}$ and $Y = \{9(n-1) : n \in \mathbb{N}\}$, then precisely

1) $X \subset Y$ 2) $X \subseteq Y$
3) $X \supset Y$ 4) $X \supset Y$

32. If $A \cup B = \phi$, then

1) $A = \phi$ or $B = \phi$ 2) $A = \phi$ and $B = \phi$
3) $A = \phi$ and $B \neq \phi$ 4) $A \neq \phi$ and $B = \phi$

33. If A has 3 elements and B has 6 elements, then the minimum number of elements in $A \cup B$ is

1) 3 2) 6
3) 9 4) 18

34. If A and B are two sets such that $n(A) = p$, $n(B) = q$ and $n(A \cap B) = r$, then $n(A \cup B)$ is equal to

1) $p + q$ 2) $p + q + r$
3) $p + q - r$ 4) $p - q - r$

35. Consider the following statements : For any two sets A, B

I. $(A - B) \cup B = A$ II. $(A - B) \cup A = A$
III. $(A - B) \cap B = \phi$ IV. $A \cup B \Rightarrow A \cap B = B$

Of these statements

1) I, II and III are correct
2) II, III and IV are correct
3) I, III and IV are correct
4) I, II and IV are correct

36. In a group of 52 persons, 16 drink tea but not coffee and 33 drink tea. How many drink coffee but not tea?

1) 3 2) 7 3) 17 4) 19

37. In a class consisting of 100 students, 20 know English, and 20 do not know Hindi and 10 know neither English nor Hindi. The number of students knowing both English and Hindi is

1) 5 2) 10 3) 15 4) 20

38. The function $f: \mathbb{N} \rightarrow \mathbb{N} : f(x) = 2x$, where $\mathbb{N}$ is the set of natural numbers is

1) one-one and onto
2) one-one but not onto
3) onto but not one-one
4) neither one-one nor onto

39. The range of the function $f(x) = a \cos(bx + c) + d$, $a > 0$ is

1) $[-a, a]$ 2) $[-a - d, a - d]$
3) $[-d - a, d + a]$ 4) $[d - a, d + a]$

40. The number 111, 111, 111, 111 is divisible by

1) 9 and 11 2) 5 and 11
3) 5 and 9 4) 3 and 11

41. A number when divided by 779 gives 47 as remainder. On dividing the same number by 19 what would be the remainder?

1) 0 2) 3 3) 9 4) 10

42. Consider the following statements :

Assertion (A) : The square of any integer is either divisible by 4 or leaves the remainder 1 when divided by 4.

Reason (R) : Since the square of any integer is of the form $4q^2 + 4q + 1$.

Of these statements,

1) both (A) and (R) are true and (R) is the correct explanation of (A)
2) both (A) and (R) are true but (R) is not the correct explanation of (A)
3) (A) is true and (R) is false
4) (A) is false and (R) is true

43. Consider the following statements :

Assertion (A) : If a number x divides the product yz and if x divides y , then x must divide z .

Reason (R) : If x is prime to each of y and z , then x is prime to the product yz .

Of these statements,

- 1) both (A) and (R) are true and (R) is the correct explanation of (A)
- 2) both (A) and (R) are true but (R) is not the correct explanation of (A)
- 3) (A) is true and (R) is false
- 4) (A) is false and (R) is true

44. The set of negative integers is closed with respect to

- 1) addition
- 2) multiplication
- 3) division
- 4) both (1) and (2)

45. The expression $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)}$ for any natural number n is

- 1) always greater than 1
- 2) always less than 1
- 3) always equal to 1
- 4) not define

46. A boy was asked to multiply a certain number by 25. He multiplied it by 52 and got his answer more than the correct one by 324. The number to be multiplied was

- 1) 12
- 2) 15
- 3) 25
- 4) 52

47. The value of $(51+52+53+\dots+100)$ is

- 1) 2525
- 2) 2975
- 3) 3225
- 4) 3775

48. If three numbers a, b, c are in A.P., then $a^2(b+c), b^2(c+a)$ and $c^2(a+b)$ are in

- 1) A.P.
- 2) G.P.
- 3) H.P.
- 4) A.G.P.

49. The value of $9+9 \times 9-9+9$ is

- 1) 17
- 2) 90
- 3) 9
- 4) 18

50. If the area of a square of side x is 7, what is the area of the square of side $3x$?

- 1) 49
- 2) 56
- 3) 63
- 4) 21

For questions (51-54) use the following :

While researching her family tree, Revathi discovered the following sketchy information about four of her ancestors:

Rajan was born before Jagan was born; Susheela died before Rajan died; Hamsa died after Jagan, Rajan, and Susheela died

51. Which of the following must be true?

- 1) Jagan was born after Hamsa was born
- 2) Hamsa was born after Susheela, Jagan, Rajan were born
- 3) Susheela was born before Jagan died
- 4) Susheela was born before Rajan died

52. If Hamsa was born before Rajan was born, which of the following must be true?

- I. Hamsa lived longer than Rajan
- II. Hamsa lived longer than Jagan
- III. Hamsa lived longer than Susheela

1. I only
2. III only
- 3) I and II only
- 4) II and III only

53. If Jagan died before Rajan died and Rajan died before Hamsa was born which of the following must be true?

- 1) Susheela was not alive during Rajan's lifetime
- 2) Jagan was not alive during Hamsa's lifetime
- 3) Susheela was not alive during Jagan's lifetime
- 4) Jagan was born and died during Susheela's lifetime

54. If Hamsa was born after Rajan died, which of the following could be true?

- 1) Jagan was Rajan's father
- 2) Rajan was Susheela's father
- 3) Hamsa was Susheela's mother
- 4) Hamsa was Rajan's wife

For questions (55-60) use the following :

During its manufacture any widget enters an assembly line having exactly seven consecutively numbered work stations (1 through 7). Each station is operated by either one human worker or by one robot. Exactly three of the operators K, L, M are human workers. Exactly two of the operators R-1 and R-2 are robots and each of the robots operates exactly two consecutively numbered stations.

An entering widget moves through stations 1 to 7 in that order. A flaw in a widget can occur anywhere on the assembly line. A flaw can be detected at any station

operated by a human worker. But flaws cannot be detected in station 1. Although only human operators can detect flaws in widgets, they do not necessarily detect every flaw.

55. Which of the following is a possible set of assignments of operators to stations?

	1	2	3	4	5	6	7
1)	R-2	R-2	L	K	M	R-1	R-1
2)	R-1	R-2	M	R-1	R-2	K	L
3)	M	R-1	R-1	K	R-2	L	R-2
4)	L	M	R-2	R-1	R-2	R-1	K

56. If K is assigned to station 7 and L is assigned to station 4, which of the following must be true?

- 1) K but not L could detect a flaw in a widget on the assembly line
- 2) M is assigned to either station 1 or 3
- 3) R-1 is assigned to either station 3 or 5
- 4) R-2 is assigned to stations 5 or 6

57. If R-2 causes a flaw in a widget at station 4, and the flaw is detected and the flawed widget is removed from the assembly line at station 6, L could operate in any station except

- 1) 1
- 2) 2
- 3) 3
- 4) 5

58. If M operates station 6, which of the following must be true?

- 1) K operates either station 1 or 7
- 2) Either K or L operates an even-numbered station
- 3) K and L operate stations that are immediately adjacent to each other
- 4) L operates an odd-numbered station

59. If R-1 operates station 3 and K is the only operator on the assembly line who is in a position to detect flaws in widgets that are created by R-1, which of the following must be true?

- 1) R-1 operates station 4
- 2) R-2 operates station 5
- 3) K operates station 4
- 4) L operates station 1

60. Under an additional rule stipulating that R-1 must operate station 3 or 4 but cannot operate both, and that R-2 must operate station 5 or 6

but cannot operate both, any would be possible robot assignment except

- 1) R-1 is assigned to station 2 and R-2 is assigned to station 4
- 2) R-1 is assigned to station 4 and R-2 is assigned to station 5
- 3) R-1 is assigned to station 3 and R-2 is assigned to station 5
- 4) R-1 is assigned to station 3 and R-2 is assigned to station 6

For questions (61–66) use the following :

The manager of a radio program is going to feature six vocalists – M, N, P, Q, R, and S on her half-hour radio show during the course of one week. She will feature one vocalist on the show each day from Monday to Saturday. The manager must schedule the vocalist according to the following conditions :

M must be featured earlier in the week than R.

P must be featured on Tuesday.

Q must be featured on the day immediately before or immediately after the day on which N is featured

61. If N is to be featured on Thursday, the earliest day on which R can be featured is

- 1) Monday
- 2) Wednesday
- 3) Friday
- 4) Saturday

62. If S is to be featured on Friday, M must be featured on

- 1) Monday
- 2) Wednesday
- 3) Thursday
- 4) Saturday

63. If Q is to be featured on Thursday, the latest day on which M can be featured is

- 1) Monday
- 2) Wednesday
- 3) Friday
- 4) Saturday

64. Which of the following vocalists can be featured on Monday?

- 1) N
- 2) Q
- 3) R
- 4) S

65. If S is to be featured on Thursday, which of the following is true?

- 1) M must be featured on Wednesday
- 2) N must be featured on Saturday
- 3) Q must be featured exactly two days after R is featured
- 4) R must be featured on Wednesday

66. If S is to be featured on Friday, what is the total number of acceptable schedules available to the manager?

1) 1 2) 2
3) 3 4) 4

Use the following for Questions (67–70) :

Distribution of SPCA registered dogs for two districts:

North Arcot :

Total number of registered dogs is 30,000. Out of which 40% is pure breed. Out of pure breed dogs, Companion is 25%, Working is 20%, Toy is 15%, Terrier is 20%, Hunting is 12% and Rare breed is 8%.

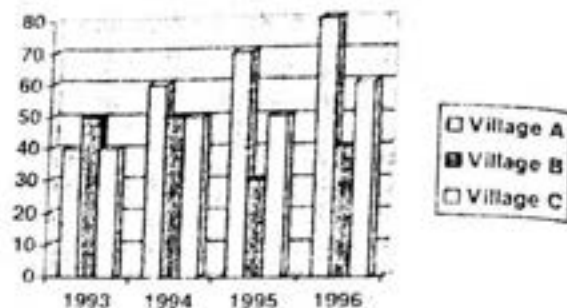
South Arcot :

Total number of registered dogs is 50,000. Out of which 50% is pure breed. Out of pure breed dogs, Companion is 20%, Working is 30%, Toy is 10%, Terrier is 25%, Hunting is 7% and Rare breed is 8%.

- 67.** How many more pure breed dogs are registered with SPCA in South Arcot than in North Arcot?
- 1) 9000 2) 11000
3) 13000 4) 15000
- 68.** For both the districts together, what is the approximate percentage of mixed breed dogs that are registered?
- 1) 24 2) 34
3) 44 4) 54
- 69.** Which of the following statements can be inferred from the data?
- I. More mixed breed dogs are registered in North Arcot than in South Arcot
II. More rare breed dogs are registered in North Arcot than in South Arcot
III. In South Arcot, companion dogs that are registered are twice as many as toy dogs.
- 1) I only 2) I and III only
3) III only 4) I and II only
- 70.** How many more terriers are registered in South Arcot than in North Arcot?
- 1) 2850 2) 3850
3) 2400 4) 6250

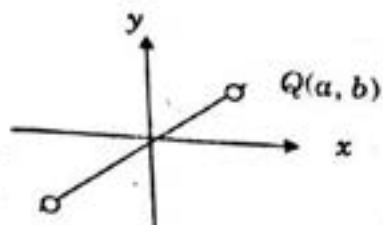
Use the following data for questions (71–75)

District Z comprises of three villages, A, B, C. The population trend in each of these 3 villages is shown below:
(Population in thousands)



- 71.** In which year did village C show the highest growth in population?
- 1) 1993 2) 1994
3) 1995 4) 1996
- 72.** In which year was the proportion of village A population the highest in district Z's total population?
- 1) 1993 2) 1994
3) 1995 4) 1996
- 73.** In 1994, the ratio of Men to Women in district Z was 1 : 1 and in 1995, the ratio was 2 : 3. Then what was the percentage increase in population of women from 1994 to 1995?
- 1) 33 2) 66
3) 20 4) 12.5
- 74.** If increase in population of villages can only be due to new births and decrease in population can only be due to deaths then number of new births in 1996 is
- 1) 20,000 2) 30,000
3) 40,000 4) cannot be determined
- 75.** The ratio of population of village A to the population of village B is the highest in which year?
- 1) 1993 2) 1994
3) 1995 4) 1996

76.

 $P(-6, -5)$ 

In the above diagram, a is _____ b .

- 1) greater than
- 2) equal to
- 3) less than
- 4) cannot be determined

77. If a unbiased coin is tossed five times. What is the probability of getting atleast one tail?

- 1) $1/5$
- 2) $1/4$
- 3) $27/32$
- 4) $31/32$

78. Three faces of a fair die are yellow, two faces red and one blue. The die is tossed three times. The probability that the colours yellow, red and blue appear in the first, second and third tosses respectively is

- 1) $1/8$
- 2) $125/216$
- 3) $1/36$
- 4) $1/9$

79. A class has 12 men and 4 women. Suppose three students are selected at random from the class. What is the probability that they are all men?

- 1) $11/28$
- 2) $165/512$
- 3) $3/512$
- 4) $13/28$

80. In a sack there were exactly 48 marbles, each of which is either red, black or yellow. The probability of randomly selecting a red marble from the sack is $5/8$, and the probability of randomly selecting a black marble from the sack is $1/8$. How many yellow marbles are in the sack?

- 1) 6
- 2) 12
- 3) 16
- 4) 24

81. A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both win a prize. What is the probability that they will not win a prize in a single trial?

- 1) $1/625$
- 2) $1/25$
- 3) $2/25$
- 4) $24/25$

82. A square is inscribed in a circle of area 32π sq. units. Find the side of the square?

- 1) 8
- 2) 4
- 3) 16
- 4) 2

83. Three circles with centers O, P, Q and radii 5.4 cm, 5.9 cm, 6.3 cm respectively are touching each other internally at a common point C. What is the length of PQ?

- 1) 0.5 cm
- 2) 0.4 cm
- 3) 0.9 cm
- 4) 1.8 cm

84. The diameter of a roller, 120 cm long, is 84 cm. If it takes 500 revolutions to level a playground, determine the cost of leveling at the rate of Rs.3 per sq. metre?

- 1) Rs. 4752
- 2) Rs. 4572
- 3) Rs. 4323
- 4) Rs. 4233

For questions (85–89), find out which does not belong to the group.

85. 1) Logic Programming
2) Procedural Programming
3) Object-oriented Programming
4) Linear Programming

86. 1) Parallel systems
2) Arithmetic and logical systems
3) Distributed systems
4) Client-Server systems

87. 1) JAVA
2) C#
3) Smalltalk
4) FORTRAN

88. 1) PHP
2) XML
3) XHTML
4) Applets

89. 1) Assembler
2) Linker
3) Spread Sheet
4) Macro processor

90. Structured Programming concept emerged due to

- 1) Donald E. Knuth
- 2) Edsger Dijkstra
- 3) John von Neumann
- 4) Alan Turing

91. Spaghetti code refers to a program that

- 1) does information hiding
- 2) has too many branch instructions
- 3) modifies itself during execution
- 4) goes into infinite loop

92. The program that converts a high-level language program into a machine language program is a

- 1) Linker
- 2) Loader
- 3) Assembler
- 4) Compiler

93. The protocol that allows a user on one computer on the Internet to log on to another computer on the Internet is

- 1) Transmission Control Protocol
- 2) Internet Protocol
- 3) Telnet
- 4) Hypertext Transfer Protocol

94. Web is a collection of

- 1) software and protocols on computers on the Internet
- 2) computers communicating with each other
- 3) programs stored in Read-only memory
- 4) interconnecting devices within a parallel computer

95. Browsers are programs that run on

- 1) Server machines
- 2) Client machines
- 3) Printers
- 4) Copiers

96. Which one is used to identify document resources on the Internet?

- 1) Universal Identifier
- 2) Uniform Resource Locator
- 3) Resource Identifier
- 4) Universal Document Locator

97. Fully qualified domain names are translated to IP address by

- 1) Translation Server
- 2) Proxy Server
- 3) Name Server
- 4) IP Server

98. XHTML is used to describe

- 1) computations
- 2) the form and layout of documents
- 3) communications
- 4) location / address

99. Waterfall model is

- 1) computational model
- 2) software development life cycle model
- 3) communication model
- 4) software testing model

100. Blackbox Testing is testing a program

- 1) knowing how the program works
- 2) without knowing how a program works
- 3) without knowing the inputs
- 4) without knowing the outputs

MCA 2008 - ANSWERS

1	3	2	1	3	3	4	1	5	3	6	3	7	2	8	4	9	4	10	1
11	4	12	4	13	*	14	2	15	4	16	4	17	3	18	4	19	2	20	4
21	2	22	1	23	2	24	1	25	3	26	2	27	4	28	1	29	3	30	4
31	1	32	2	33	2	34	3	35	2	36	4	37	2	38	2	39	4	40	4
41	3	42	2	43	4	44	1	45	2	46	1	47	4	48	1	49	2	50	3
51	4	52	3	53	2	54	2	55	1	56	2	57	2	58	4	59	3	60	2
61	2	62	1	63	2	64	4	65	4	66	2	67	3	68	4	69	4	70	2
71	2	72	3	73	4	74	2	75	3	76	1	77	4	78	3	79	1	80	2
81	4	82	1	83	2	84	1	85	4	86	2	87	4	88	1	89	4	90	2
91	2	92	4	93	3	94	1	95	1	96	4	97	3	98	2	99	2	100	2